What is ElasticSearch?

A real-time, distributed search and analytics engine, ElasticSearch is used for full text search, structured search, analytics, and all three in combination. In short, it helps us to make sense of all the data that's being hurled at us every day.

Every year, billions of gigabytes of data are generated in various forms. We are swimming in an ocean of data. For many years, we have focused on generating and storing data in different structured forms. This data is not useful to us unless we can get some insights into it using analytics. We need tools to explore the data, and present it in a reader-friendly manner or in visual forms, so as to make sense out of it. It is rightly said that a picture is worth a thousand words.

There are many tools in the market for analytics, and ElasticSearch stands out among them. It is an open source tool with which we can analyse data. According to its website, ElasticSearch is a real-time, distributed search and analytics engine. It is used for full text search, structured search, analytics and all three in combination.

ElasticSearch is built on Apache Lucene, which is an open source library for high-performance, full-featured text search [*https://lucene.apache.org/core/*]. As Apache Lucene is a library, we need to do a lot of coding to integrate it with existing applications. The Apache Lucene library is built in Java; hence, the latter is required to use the former. ElasticSearch uses Apache Lucene internally for indexing and searching, but hides its complexities by providing access to the API in the form of the RESTful API, using JSON as the format for exchanging data.

ElasticSearch is a product of the Elastic Company, which was founded in 2012. The firm has also built Logstash, Kibana and Beats, all of which are open source projects. Apart from these, there are some commercial products available like Marvel, Shield, Watcher, Found, etc. More details about the products and what they do can be found at *https://www.elastic.co/products*.

For this article, we will use ElasticSearch to store, search and analyse the data. We will also use Kibana to visualise and explore the data. The remaining products are out of the scope of this article.

Well-known users of ElasticSearch

Facebook, Microsoft, NetFlix, Wikipedia, Adobe, Stackoverflow, eBay and many other companies are using ElasticSearch in various ways to explore and analyse data. More details in the form of use cases are available online at *https://www.elastic.co/use-cases*.